domain of pure abstract research into that of practical therapy. It would of course be foolhardy to attempt to apply these methods to the human body before they have been tested sufficiently.

Another possible application of artificially produced radioactive substances is in the investigation of functional disturbances of organs. Many of the organs, both of human and animal bodies, have not just one but sometimes several functions. If such an organ is injured, it is often difficult to ascertain which of its functions is impaired and which is still carried out normally. Since every function is associated with a different type of metabolism, it is possible, for instance, to introduce into the organism certain radioactive substances, which are specific for certain functions, and to observe whether or not the organism does with them what it is supposed to do. It is possible by means of this technique to distinguish the impaired functions from the unimpaired ones. A new method of medical diagnosis may develop along these lines.

IX. THE USE OF STABLE ISOTOPES

Among the rare isotopes of the various elements, a special part is played by deuterium, the hydrogen isotope of mass number 2, since the ratio of its mass to the mass of the common hydrogen isotope of mass number 1 is far greater than usually found among isotopes of other elements. For this reason, there are noticeable differences in the chemical properties of these two isotopes, which make it easy for us to detect them when they are together. Consequently heavy hydrogen can also be used as a tracer. For instance, fatty acids have been built up with heavy hydrogen, instead of the common hydrogen isotope, and these fatty acids have been introduced into an organism, and the problem studied as to how the organism makes use of them. This experiment revealed that the long-chained fatty acids settle in the liver and in the fatty tissues, whereas the short-chained ones are consumed immediately. The experiment would not have been possible with ordinary hydrogen, because such fatty acids are always present in the organism, and therefore it would have been impossible to distinguish the fatty acids deliberately inserted into the food from those previously present in the organism. Similar experiments have been performed with nitrogen of mass number 15 and oxygen of mass number 18.

In conclusion, let us mention one more application of a nuclear reaction to a branch of physics itself—to optics. With the cyclotron, the opposite of the old dream of the alchemists—not a conversion of mercury into gold, but one of gold into mercury—has been achieved. Gold is monoisotopic, in other words, it has only one stable isotope, $_{79}Au^{197}$. If a nuclear reaction is